

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0534

Olney 100 kV
August 21, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0534. The Interconnection Customer proposes a 182.9 MW wind facility interconnecting at the Olney 100 kV substation in the Ohio area. The request was received on November 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 212 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,732,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of OLNEY-BETHPAGE POWER PLANT 100.0 kV line, the ASCARTE-OLNEY 100.0 kV loads to 156.1% of its 104 MVA rating. Project DFAX: 100.0%.

The OLNEY-BETHPAGE POWER PLANT 100.0 kV is overloaded to 134.5% of its 121 MVA emergency rating for the loss of the OLNEY-ASCARTE 100.0 kV L. The project contributes 100.0% DFAX (182.9 MW) to this overload.

For the loss of BETHPAGE POWER PLANT-OLNEY SUBSTATION 100.0 kV, the ASCARTE-LAWNDALE 100.0 kV loads to 129.2% of its 115 MVA rating. Project DFAX: 100.0%.

For the loss of DEKALB-JONES CHAPEL 500.0 kV, the RUSSELL FORK-BUEBONNET 230.0 kV loads to 112.5% of its 900 MVA rating. Project DFAX: 5.5%.

CADIZ-LAWNDALE 100.0 kV: 106.0% loading (124 MVA) following the outage of the OLNEY-BETHPAGE POWER PLANT 100.0 kV; DFAX = 100.0%.

The BLUEBONNET-JONES CHAPEL 230.0 kV line is overloaded to 105.1% of its 1,085 MVA emergency rating for the loss of the ROBISON PARK-RUSSELL FORK 230.0 kV. The project contributes 5.1% DFAX (9.4 MW) to this overload.

For the loss of RICHLAND-DEKALB 500.0 kV, the DEKALB SUBSTATION 230-1 kV Transformer loads to 104.3% of its 1,082 MVA rating. Project DFAX: 6.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ASCARTE-OLNEY 100.0 kV	Rebuild line with higher-capacity conductor	\$3,116,000	100% (\$3,116,000)
OLNEY-BETHPAGE POWER PLANT 100.0 kV	Rebuild line with higher-capacity conductor	\$7,535,000	100% (\$7,535,000)
ASCARTE-LAWNDALE 100.0 kV	Rebuild line with higher-capacity conductor	\$10,164,000	100% (\$10,164,000)
BLUEBONNET-RUSSELL FORK 230.0 kV L	Rebuild line with higher-capacity conductor	\$44,909,000	6% (\$2,608,000)
CADIZ-LAWNDALE SUBSTATION 100.0 kV	Reconductor line	\$2,232,000	100% (\$2,232,000)
BLUEBONNET-JONES CHAPEL 230.0 kV	Reconductor line	\$16,933,000	4% (\$650,000)
DEKALB SUBSTATION 230-1 kV Transformer	Replace transformer with higher-rated unit	\$8,467,000	5% (\$427,000)
	Total allocated to project		\$26,732,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.